

FTC Control Hub — Mac Setup From Scratch

Install each piece, set your PATH, and get past the usual zsh snags — then push TeamCode to the REV Control Hub.

1 Check what you're working with

```
uname -m
echo $SHELL
```

`arm64` means Apple Silicon — Homebrew lives in `/opt/homebrew`. `x86_64` means Intel — it lives in `/usr/local`.

If the shell isn't `/bin/zsh`, run `chsh -s /bin/zsh` and reopen Terminal.

2 Install Homebrew

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

It asks for your Mac password and installs Apple's Command Line Tools if they're missing. This takes several minutes.

3 Put Homebrew on your PATH

The installer doesn't do this for you — which is why `brew` gives "command not found" right after installing.

Apple Silicon

```
echo 'eval "$(/opt/homebrew/bin/brew shellenv)"' >> ~/.zprofile
eval "$(/opt/homebrew/bin/brew shellenv)"
```

Intel

```
echo 'eval "$(/usr/local/bin/brew shellenv)"' >> ~/.zprofile
eval "$(/usr/local/bin/brew shellenv)"
```

Check with `brew --version`.

4 Install ADB and Android Studio

```
brew install --cask android-platform-tools
brew install --cask android-studio
```

Homebrew links `adb` into its own bin folder, so once step 3 works, `adb` is on PATH too. Check with:

```
which adb
adb version
```

5 If zsh is the problem

Symptom	Fix
command not found: brew or adb in a new window	Step 3 didn't save. Run <code>cat ~/.zprofile</code> to confirm the line is there, then reopen Terminal.
no such file: ~/.zprofile or ~/.zshrc	Run <code>touch ~/.zprofile ~/.zshrc</code> , then redo step 3.
permission denied writing to those files	<code>sudo chown \$USER ~/.zprofile ~/.zshrc</code>
Errors every time Terminal opens	A bad line in <code>~/.zshrc</code> . Run <code>mv ~/.zshrc ~/.zshrc.bak</code> and reopen Terminal.
Even <code>ls</code> says "command not found"	PATH got wiped. Run <code>export PATH="/usr/bin:/bin:/usr/sbin:/sbin:/opt/homebrew/bin"</code> , then fix or move <code>~/.zshrc</code> as above.
compinit: insecure directories	<code>compaudit xargs chmod g-w</code>
Android Studio's terminal can't find adb	Quit and reopen Android Studio so it picks up the new PATH.

6 Get the FTC project (still on normal internet)

Download the ZIP from GitHub: **FIRST-Tech-Challenge/FtcRobotController**.

Open the folder in Android Studio and let the Gradle sync finish.

Run **Build > Make Project** once, so everything is downloaded **before** you join the hub's Wi-Fi, which has no internet.

7 Connect to the Control Hub

Power the hub from the robot battery and wait for the **green LED**.

USB-C

Plug in a **data-capable** cable, click **Allow** if macOS asks, then run `adb devices`. One device should be listed.

Wi-Fi

Join the hub's network (the name is on the hub's sticker; the factory password is `password`), then run:

```
adb connect 192.168.43.1:5555
```

It should reply `connected to 192.168.43.1:5555`.

8 Push code

In the Android Studio toolbar, choose **TeamCode** and the **REV Control Hub** as the device, then click **Run ►**. The Robot Controller app restarts with your code.

► One conflict to watch for

Android Studio also downloads its own copy of ADB. If you see "**adb server version doesn't match this client**" repeating, the two copies are different versions. Run `brew upgrade --cask android-platform-tools` and update **Platform-Tools** in Android Studio's SDK Manager so they match, then run `adb kill-server`.